3 **(a)** Two light identical springs are joined in series and a 120 g bar magnet is hung from the lower end as shown in Fig. 3.1. The effective spring constant of the combined springs is 11 N m^{-1}.

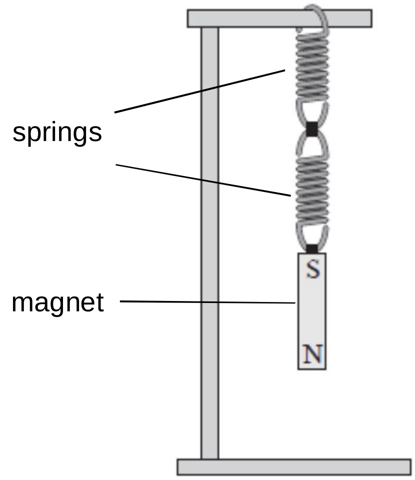

Fig. 3.1

The bar magnet is displaced a small distance vertically downward from its equilibrium position and released. It then oscillates in simple harmonic motion.

(i) By considering the forces on the magnet at its equilibrium position, and also at a small displacement x downward from its equilibrium position, show that the oscillation is a simple harmonic motion where the acceleration a is given by the equation

$$a = -92\,x$$

[4]

Let downwards be positive.

Let e be the extension of the spring when the magnet is at its equilibrium position.

At the equilibrium position, **M1**
$ke = mg$ ---------- (1)

At a displacement x downward from its equilibrium position, **M1**
$mg – k(x + e) = ma$
$mg – kx – ke = ma$ -------- (2)

Sub (1) into (2):
$-kx = ma$ **M1**
$a = -(k/m)x$
$a = -[11/(120 \times 10^{-3})]x$ **M1**
$a = -91.667x$
$a = -92x$ (shown) **A0**

(ii) Hence calculate the frequency at which the magnet oscillates.

frequency = ... Hz [2]

$a = -92x$
Comparing with the SHM equation, $a = -\omega^2 x$,
$\omega^2 = 92$ **M1**
$\omega = \sqrt{92}$
$2\pi f = \sqrt{92}$
$f = 1.5266 = 1.5$ Hz **A1**

(b) Two systems in part **(a)** are now suspended with the North pole of their magnets located inside a coil of wire. The two coils are connected together with conducting leads as shown in Fig. 3.2.

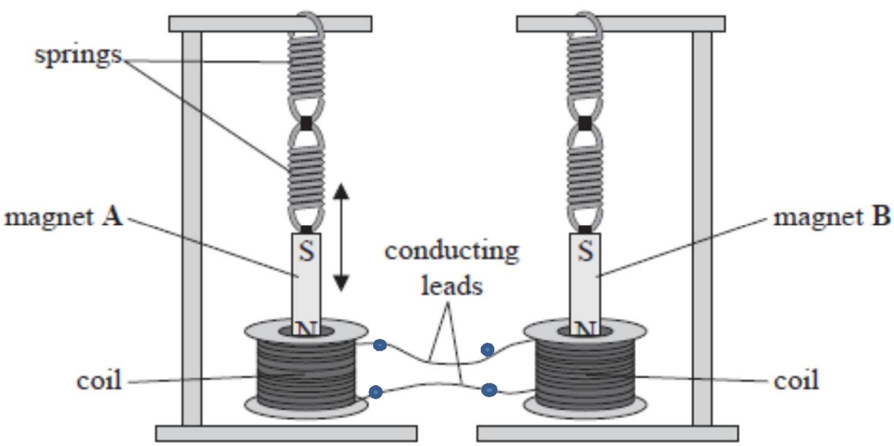

Fig. 3.2

Magnet A is displaced so that it oscillates vertically. The North pole of magnet A moves into and out of the coil of wire with simple harmonic motion. As this motion continues, magnet B starts to oscillate. The amplitude of oscillation of magnet B increases over time.

Explain why magnet B starts to oscillate with an increasing amplitude.

-- [5]

As magnet A moves in and out of the coil, its coil experiences a change of magnetic flux density and hence a change in magnetic flux linkage. **B1**

By Faraday's law, this change in magnetic flux linkage induces an electromotive force in the coil which then causes a current in both coils (since the coils are connected to each other). **B1**

The alternating current in the second coil creates a periodic force on magnet B and driving it into a forced oscillation. **B1**

Since both spring-magnet systems have the same natural frequency of oscillation, **B1**

resonance occurs and thus there is a maximum transfer of energy from magnet A to magnet B and magnet B oscillates with increasing amplitude. **B1**